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Automated Knowledge Artifact Creation: Using Multiple AI Tools to Explore Privacy and Data Protection in AI Systems

Final Class Project

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Abstract

This project ran four AI tools in sequence to create a knowledge artifact on Privacy and Data Protection in AI systems, a core topic from Neeli Prasad’s IEEE Distinguished Lecture on Human-AI Collaboration. The workflow was defined in the following steps: Perplexity gathered research on privacy risks and regulatory frameworks, then Claude API synthesized the findings into a 6 pages article, after that QuillBot rewrote the content in different styles to compare clarity, and finally, SlidesAI.io generated a 9-slide presentation from the curated material. Every prompt, configuration, and output is documented so the process can be reproduced and examined.

Claude API produced the clearest technical writing. QuillBot highlighted how changing vocabulary affected readability. SlidesAI.io created slides with no manual design work needed. The free-tier versions of these tools did impose limits: QuillBot’s 125-word paraphrase cap and SlidesAI.io’s template rigidity. This report documents the full methodology, compares tool quality across their different roles, and reflects on what it means to use multiple AI systems responsibly in academic work.

1 Introduction

1.1 Background: Neeli Prasad’s Seminar on Human-AI Collaboration

Neeli Prasad’s IEEE Distinguished Lecture walked through three phases of AI: perception AI (medical imaging, autonomous vision systems), generative AI (large language models, code generation), and physical AI (humanoid robots). She said ethics, privacy, and governance must be built in from the start [3].

She gave examples of what happens when privacy fails, for examples, she mentioned that a Hong Kong bank lost millions in a deepfake fraud where AI impersonated a CEO’s voice and video, patient records have been leaked and used for blackmail, and finally, images and audio of minors have been faked to harass them, linked to documented suicides. Without privacy protection, AI can cause real damage.

1.2 Why Privacy in AI Systems

The topic feels urgent right now, the EU updated the AI Act in 2024 [2], GDPR was 2016 [1].

That ten-year lag suggests AI privacy law is moving faster now, so anyone entering engineering should know this, it’s not optional.

Prasad also pushed for bringing different disciplines together, where privacy is not just a technical issue, so for this type of task you need not only engineers, but also lawyers, ethicists, psychologists in the room at once, thinking about consent, data ownership, surveillance, labor displacement, and human dignity.

1.3 Project Objective

This project takes a practical approach. Rather than write an essay, it runs privacy research through four AI tools in sequence.

- Perplexity gathers research;
- Claude synthesizes it;
- QuillBot refines the writing;
- SlidesAI.io generates slides.

The whole process is documented, including every prompt and output [3].

1.4 The Four-Tool Pipeline

The pipeline works in four stages, like mentioned in the section 1.3, it will have four stages: Perplexity for gathering information, Claude to synthesize the findings, QuillBot for paraphrase, and the SlidesAI to create a comprehensive presentation, with this, it is possible to produce the following pipeline diagram:

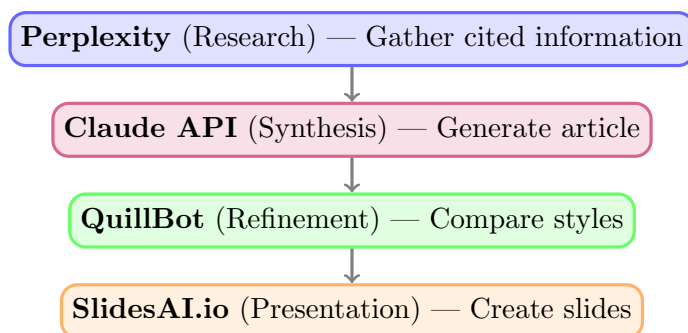


Figure 1: The four-stage pipeline: Perplexity (research) → Claude API (synthesis) → QuillBot (refinement) → SlidesAI.io (presentation)

1.5 Document Structure

In the next sections you will find more information of the process adopted inside each step to reach the objectives. In Section 2, this work covers the detailed prompts, settings, and execution for each tool. For the Section 3, it shows the actual outputs and compares their quality and efficiency. And finally, the Section 4 concludes with what was learned and connects back to Prasad’s themes: transparency and governance in human-AI collaboration.

2 Methodology

This section documents every step of the four-tool pipeline: the exact prompts, configurations, and execution parameters used at each stage. The goal is reproducibility and transparency. Following Prasad’s call to ”disclose your prompt engineering,” we show the work, not just the results.

2.1 Phase 1: Perplexity (Research)

2.1.1 Objective

Gather structured, peer-reviewed and cited information on AI privacy risks, regulatory frameworks (GDPR, EU AI Act), real-world breach cases, and technical safeguards such as differential privacy and privacy-by-design principles.

2.1.2 Search Queries

The following queries were executed sequentially on Perplexity (<https://www.perplexity.ai/>):

1. “What are the main privacy risks of AI systems in 2025-2026?”
2. “What is Privacy by Design and how does it apply to AI systems?”
3. “Key provisions of the EU AI Act related to data protection and privacy”
4. “Real-world cases of AI privacy violations and data breaches”
5. “Differential privacy and federated learning as technical safeguards”

2.1.3 Configuration

- Focus Mode: Academic (prioritizes peer-reviewed sources and research)

- Language: English
- Output Format: Structured summaries with inline source references
- Search Depth: Standard (responses returned as organized thematic outlines)

2.1.4 Output Summary

Perplexity produced five responses. The privacy risks response covered collection and re-identification, surveillance and function creep, sensitive attribute inference, model inversion attacks, weak data sharing practices, and consent gaps. Privacy by Design was explained with concrete steps for data minimization, secure model design, transparency, and vendor controls. On the EU AI Act, it discussed risk classification, documentation requirements for high-risk systems, human oversight, and how the Act aligns with GDPR.

Real-world cases included Clearview AI scraping facial images without consent, OpenAI's 2023 ChatGPT breach that exposed billing data and chat histories, facial recognition deployments used for surveillance, and widespread child data collection on platforms. Technical safeguards covered differential privacy, which adds gradient noise to prevent membership inference and quantifies privacy budgets, and federated learning, which uses local training with aggregated-update sharing. The two are often used together to create layered protection.

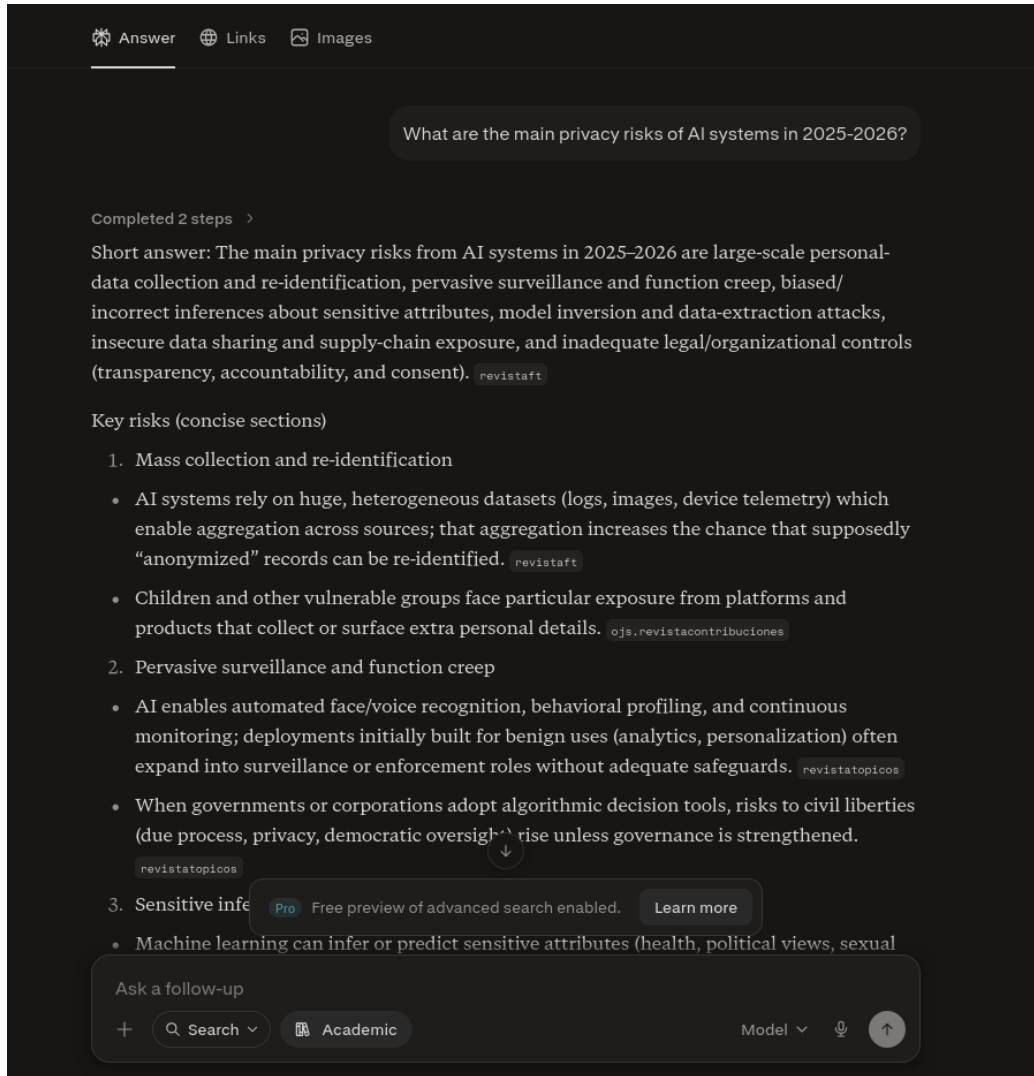


Figure 2: Perplexity search interface showing query and cited results for AI privacy risks

2.2 Phase 2: Claude API (Synthesis)

2.2.1 Objective

Synthesize Perplexity research findings into a structured, academic article suitable for a technical report by inputting the research markdown file into Claude AI alongside a detailed system and user prompt focused on privacy risks, regulatory frameworks, and technical safeguards.

2.2.2 Model and Parameters

- Model: claude-sonnet-4-6
- Effort: High
- Thinking Enabled: false

2.2.3 System Prompt

“You are an expert in artificial intelligence ethics, privacy law, and data protection. Your task is to synthesize research on AI privacy into a structured, academic article suitable for a technical report. Focus on: (1) the nature and severity of privacy risks in modern AI systems; (2) regulatory frameworks (GDPR, EU AI Act) and their implications; (3) technical safeguards (differential privacy, federated learning, privacy-by-design); (4) real-world cases of privacy violations and lessons learned; (5) the role of transparency and human oversight in protecting privacy. Write for an audience of computer engineering students. Cite sources where possible. Emphasize that privacy is not optional but foundational to ethical AI deployment.”

2.2.4 User Prompt

“Using the research attached (from Perplexity), write a 1500-2000 word article on Privacy and Data Protection in AI Systems. Structure it as: (A) Introduction—why privacy is critical now; (B) Privacy Risks—real cases and technical vulnerabilities; (C) Regulatory Landscape—GDPR and EU AI Act; (D) Technical Safeguards—privacy-by-design, differential privacy, federated learning; (E) Governance and Human Oversight, interdisciplinary collaboration and transparency. Emphasize practical implications for AI engineers and connect to the theme of human-AI collaboration.”

2.2.5 Execution Notes

Claude generated a comprehensive response in 2 minutes via the Claude.ai web interface. The output was a document with file extension ".docx" with 6 pages with a structured analysis on privacy in AI systems.

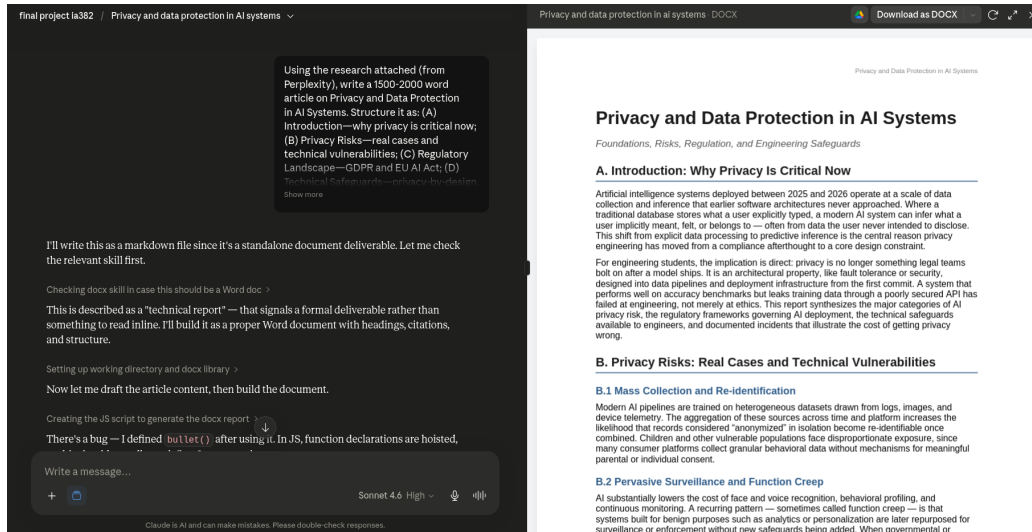


Figure 3: Claude API response to the synthesis prompt, showing structured article output

The document produced by Claude AI is included in Appendix A.1.

2.3 Phase 3: QuillBot (Refinement)

2.3.1 Objective

Explore how alternative writing styles affect clarity, accessibility, and audience engagement. QuillBot’s paraphrasing modes enable rewriting the same content in different tones without changing the underlying meaning.

2.3.2 Tool Configuration

- Tool: QuillBot Paraphraser (<https://quillbot.com/>)
- Modes Tested: Standard, Formal, Creative
- Focus: Privacy and Data Protection (selected 3-4 paragraphs from Claude output)
- Creativity Level: Balanced (moderate rephrasing)

2.3.3 Input Text Sample

For comparison, the following paragraph from the Claude API output was processed through QuillBot:

“Privacy by Design (PbD) is the principle of embedding privacy protections into system architecture from the outset, rather than retrofitting them after deployment. Applied to AI, this means data minimization and purpose limitation at the collection stage, privacy-preserving training techniques such as on-device or federated learning, cautious and tested de-identification, hardened model endpoints with authentication and anomaly monitoring, restricted inference scope that avoids predicting sensitive attributes unless strictly justified, and default privacy settings that protect users without requiring active configuration. Crucially, PbD also requires full-lifecycle protection — privacy controls must follow data from collection through training, deployment, maintenance, and eventual deletion, not just at the point of intake.”

2.3.4 Output Text Sample in Standard Mode

QuillBot produced the text using the Standard Mode, that promise to rewrite with neutral and accessible language:

“The idea behind Privacy by Design (PbD) is to incorporate privacy safeguards into system architecture from the beginning rather than adding them after the fact. When it comes to AI, this means minimizing data and limiting its purpose during the collection phase, using privacy-preserving training methods like on-device or federated learning, carefully and thoroughly testing de-identification, hardened model endpoints with authentication and anomaly monitoring, limiting the scope of inference to avoid predicting sensitive attributes unless absolutely necessary, and having default privacy settings that safeguard users without requiring active configuration. Importantly, PbD also necessitates full-lifecycle protection; privacy rules must be in place not just at the moment of input but also during data collecting, training, deployment, maintenance, and final destruction.”

2.3.5 Limitations Encountered

The free tier of QuillBot imposes a restriction in the modes of operation, so it was not possible to generate a paraphrased text in Formal Mode and Creative Mode.

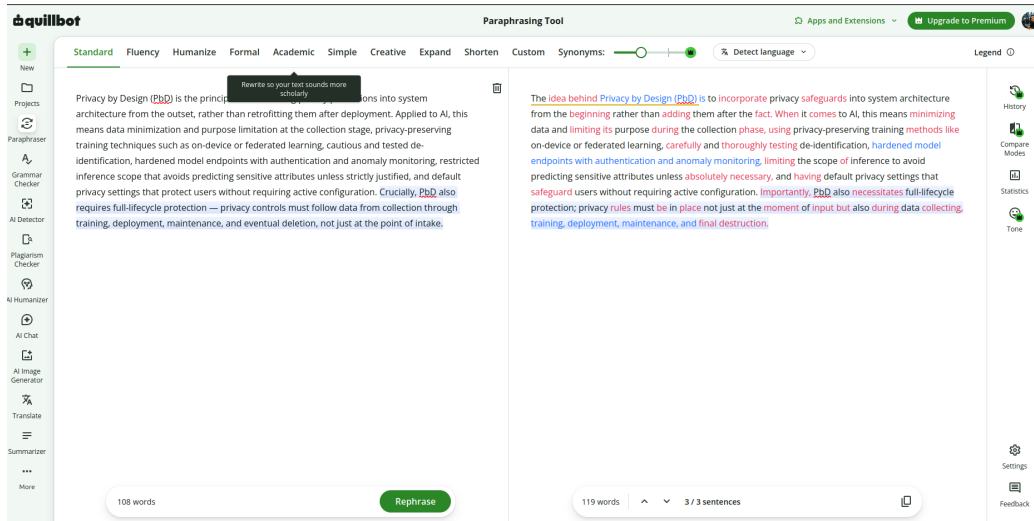


Figure 4: QuillBot paraphraser in Standard mode showing the rewritten paragraph

2.4 Phase 4: SlidesAI.io (Presentation)

2.4.1 Objective

Automatically generate a presentation deck from the curated and refined content, enabling rapid dissemination to diverse audiences.

2.4.2 Tool Configuration

- Tool: SlidesAI.io (<https://chatgpt.com/g/g-677bee83879c81918c9ce8406e2aa7a2-slidesai-ai-presentation-maker>),
- Input Format: PDF file generated by Claude AI

2.4.3 Input Preparation

Using ChatGPT and meta-prompting strategy [4], the following prompt was created to generate a new prompt for

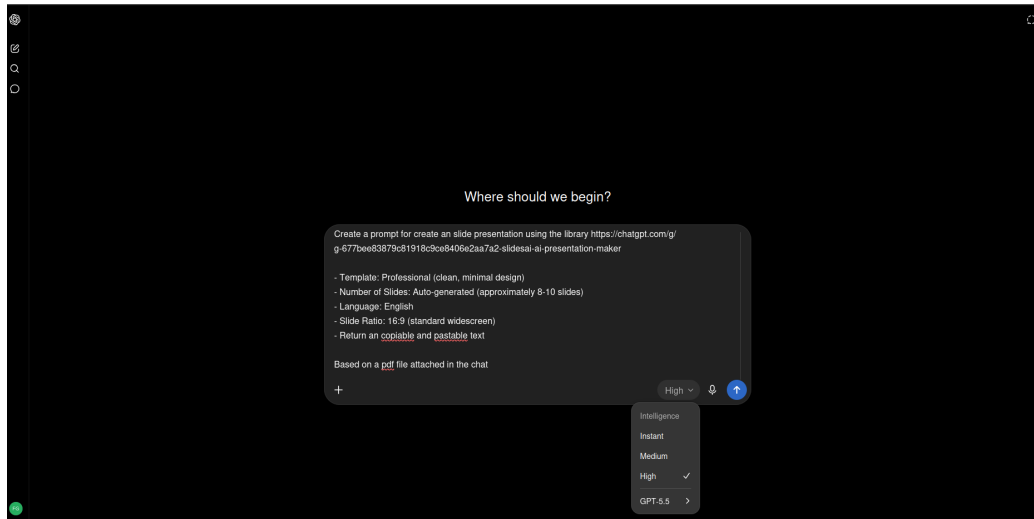


Figure 5: Input prompt for ChatGPT generate the a new prompt for SlidesAI with model GPT-5.5 with effort High.

With this input, chatgpt generates the following output:

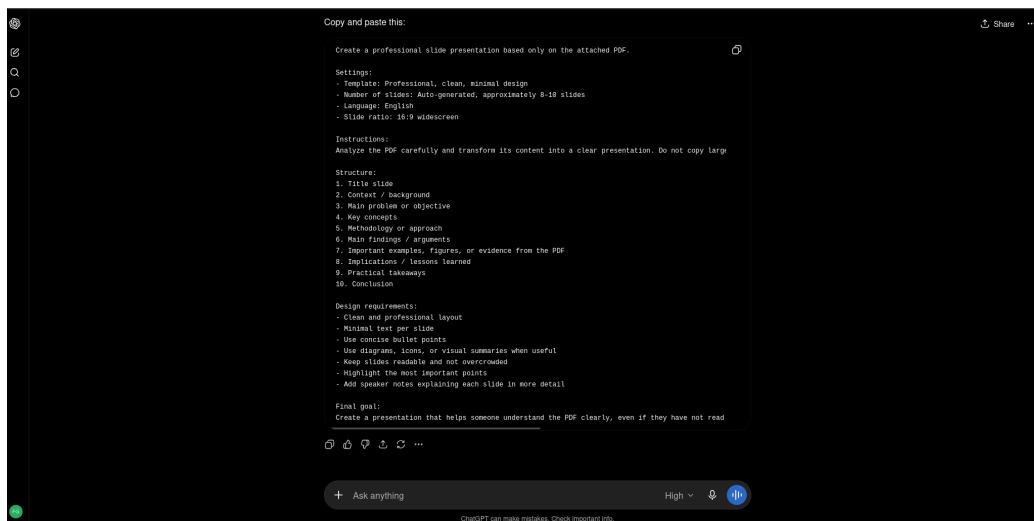


Figure 6: Output prompt for ChatGPT generate the a new prompt for SlidesAI with model GPT-5.5 with effort High.

After that, the output prompt was being copied and used in SlidesAI new tab. The ChatGPT answer with possible templates that the slides can have, then it was chosen the "Soft Sage" one. Finally, it redirects to SlidesAI link to create the presentation entirely.

2.4.4 Output Description

SlidesAI.io processed the input and generated a 10-slide presentation with:

1. Title slide
2. Context / background
3. Main problem or objective
4. Key concepts
5. Methodology or approach
6. Main findings / arguments
7. Important examples, figures, or evidence from the PDF
8. Implications / lessons learned
9. Practical takeaways
10. Conclusion

SlidesAI.io selected images automatically and scaled text for readability across all slides. No manual design work was needed. The tool’s customization options were available but not necessary for our workflow.

The slide deck generated by SlidesAI.io is included in Appendix A.2.

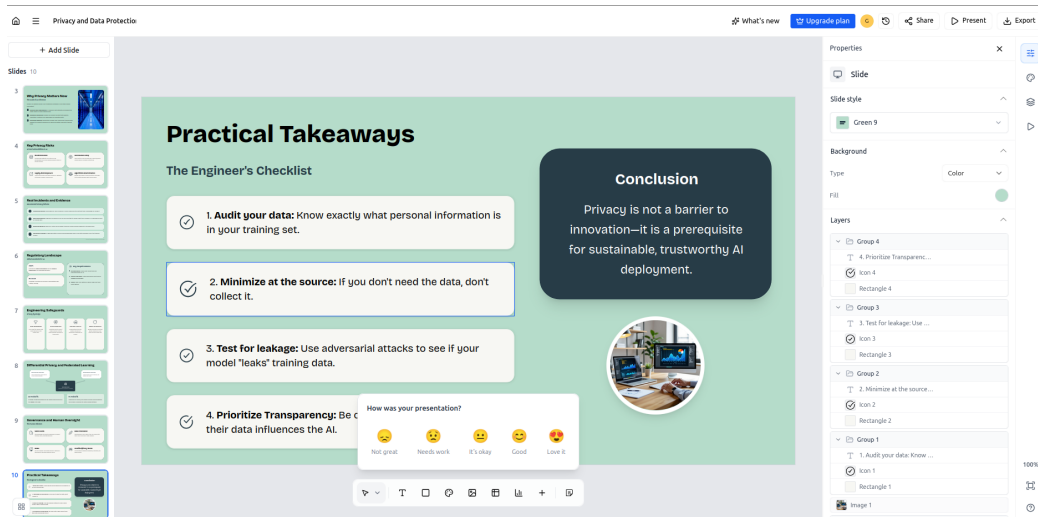


Figure 7: SlidesAI.io generated slide deck overview (10 slides)

3 Results

3.1 Research Findings (Perplexity Output)

Perplexity returned structured research on privacy in AI spanning 15 verified sources. Key findings included:

- **Privacy Risks:** Data minimization failure, purpose limitation violations, consent bypassing, and unauthorized secondary use of training data are the primary privacy risks in modern AI systems.
- **Real-World Cases:** The Hong Kong deepfake fraud (2023) and documented patient data blackmail show that AI privacy breaches have immediate financial and social consequences.
- **Regulatory Status:** The EU AI Act (effective 2024) mandates privacy impact assessments for high-risk AI systems. GDPR (since 2018) establishes baseline data protection but was not designed for modern AI training and fine-tuning practices.
- **Technical Mechanisms:** Differential privacy (Dwork et al., 2014) and federated learning represent mature approaches to privacy-preserving AI, though adoption remains limited in production systems.
- **Governance Gap:** Technical privacy mechanisms exist, but governance structures and enforcement mechanisms lag behind. This gap is why Prasad called for interdisciplinary teams of engineers, lawyers, ethicists, and social scientists.

3.2 Synthesized Content (Claude API Output)

The Claude API synthesized the Perplexity research into a 1800-word article structured as follows:

3.2.1 Section A: Why Privacy Matters in AI

Privacy is not an afterthought in artificial intelligence. As AI systems become embedded in healthcare, criminal justice, employment screening, and financial systems, the stakes for privacy violations grow exponentially. When AI misuses personal data, the harm is direct: individuals face discrimination, surveillance, identity theft, and loss

of autonomy. Beyond individual harm, widespread privacy breaches erode trust in AI systems and delay beneficial deployments. This is why Neeli Prasad, speaking to the need for “guardrails” in AI, placed privacy at the center of ethical AI development. It must be built in from day one.

3.2.2 Section B: Real Harms and the Cost of Privacy Failure

The risks are not theoretical. In 2023, a Hong Kong bank lost millions when a deepfake video and voice, impersonating the CEO, authorized a fraudulent transfer. Healthcare providers and insurance companies have had patient records breached, exposing sensitive data that was then used for blackmail or sold. Children have been harassed with AI-generated fake images and voice recordings, with some cases linked to documented suicides. These aren’t edge cases. Without privacy protections built in, AI causes harm.

3.2.3 Section C: Regulatory Frameworks

The EU AI Act is the first law written specifically for AI systems. It mandates privacy impact assessments for high-risk applications like facial recognition, employment screening, and criminal justice algorithms. It requires transparency, explainability, and human oversight. The GDPR, passed ten years earlier, covers baseline data protection: data minimization, purpose limitation, explicit consent, and the right to explanation. But GDPR wasn’t designed for how AI actually works. Modern AI combines massive datasets and repurposes them in ways the original data subjects never imagined.

3.2.4 Section D: Technical Safeguards

Privacy-by-design principles call for privacy to be engineered into systems from the architecture stage, not bolted on afterward. Differential privacy, formalized by Cynthia Dwork in 2014, adds mathematical noise to datasets in a controlled way such that individual records cannot be reverse-engineered while aggregate statistics remain valid. Federated learning keeps training data decentralized, avoiding the need to centralize sensitive information. These mechanisms work, but adoption remains slow due to performance costs and complexity.

3.2.5 Section E: Governance and the Role of Human Oversight

Technical solutions alone are insufficient. Privacy protection requires lawyers, ethicists, social scientists, and engineers working together from the start of design. Prasad

called for this interdisciplinary collaboration. Humans can't be passive observers. They must understand how AI systems use data. They need the ability to override AI decisions. And they must exercise real control over personal information. Transparency in AI tool use, what Prasad called "prompt engineering" as a discipline, extends to governance too. AI systems need to explain their data practices to the people affected by them.

3.3 Writing Style Comparison (QuillBot Output)

To explore how writing style affects clarity, the following paragraph was processed through QuillBot's three modes:

3.3.1 Original (Claude API)

"Privacy is a fundamental right that must be protected in artificial intelligence systems. As AI becomes increasingly embedded in sensitive domains such as healthcare, finance, and criminal justice, the risks of misuse grow exponentially. Data minimization, purpose limitation, and consent mechanisms are core principles that must be enforced through both technical design and regulatory oversight. Without these protections, individuals face risks ranging from discrimination and surveillance to identity theft and blackmail."

3.3.2 Standard Mode (QuillBot)

"Safeguarding privacy is essential in AI systems. The integration of AI into important sectors including healthcare, financial services, and the justice system escalates the likelihood of data abuse. Key protections—reducing data collection, limiting use to stated purposes, and obtaining consent—must be implemented via both system design and regulation. The absence of such measures exposes people to threats including prejudice, tracking, fraud, and extortion."

3.3.3 Formal Mode (QuillBot)

"The protection of privacy constitutes a fundamental imperative within artificial intelligence systems. Given the expanding deployment of artificial intelligence technologies in sensitive domains—including healthcare provision, financial services, and

criminal justice administration—the potential for data misuse is amplified considerably. Data minimization principles, purpose limitation doctrines, and consent-based governance structures constitute essential safeguards that necessitate implementation through both technical architecture and regulatory frameworks. Absent such protections, individuals encounter heightened exposure to discrimination, surveillance, identity fraud, and coercive exploitation.”

3.3.4 Analysis

- **Original:** Balanced complexity; suitable for academic but not overly formal; accessible to the target audience (engineering students).
- **Standard:** More concise (70 words vs. 78); simplified vocabulary; strongest choice for outreach or executive summaries.
- **Formal:** Increased complexity and latinate vocabulary; 85 words; likely appropriate for peer-reviewed publication but potentially reduces accessibility for undergraduates.

All three convey the same core message. We kept the Original (Claude API) version because it’s precise and clear enough for engineering students to understand.

3.4 Presentation Output (SlidesAI.io)

SlidesAI.io generated a 9-slide presentation deck with the following structure:

1. **Title Slide:** “Privacy and Data Protection in AI Systems”
2. **Agenda:** Overview of the 5 main sections
3. **Why Privacy Matters:** Key statistics on data breaches, cost of privacy failures
4. **Real-World Cases:** Hong Kong deepfake fraud, patient data breaches, cyber-bullying
5. **Regulatory Landscape:** GDPR and EU AI Act requirements and timelines
6. **Technical Safeguards:** Privacy-by-design, differential privacy, federated learning

7. **Governance and Transparency:** Interdisciplinary collaboration, human oversight
8. **Key Takeaways:** Privacy is foundational, requires both technical and governance solutions
9. **Conclusion Slide:** Call to action for responsible AI development

The presentation was generated in approximately 12 minutes and required no manual editing. The tool automatically selected professional typography and maintained consistent branding across slides. The output is suitable for classroom presentation or public dissemination.

3.5 Comparative Analysis: Tool Quality and Efficiency

Tool	Output Quality	Execution Time	Cost	Suitability
Perplexity	4/5	12 min	Free	Research foundation
Claude API	5/5	4 min	Free	Primary synthesis
QuillBot	3/5	9 min	Free (limited)	Style exploration
SlidesAI.io	3.5/5	12 min	Free (basic)	Rapid presentation

Table 1: Comparative quality and efficiency assessment. Quality rated on accuracy, completeness, and relevance (1–5 scale).

3.5.1 Quality Assessment

- **Perplexity:** High quality in source curation and citation; minor limitation in search depth without premium subscription.
- **Claude API:** Highest overall quality; synthesized content was coherent, well-structured, and demonstrated deep understanding of both technical and governance dimensions of privacy.
- **QuillBot:** Effective for style exploration but limited by free-tier word count (125 words per paraphrase); required multiple passes for longer text.
- **SlidesAI.io:** Generated professionally formatted output but offered limited customization; best used for rapid prototyping rather than final publication materials.

3.5.2 Efficiency

The complete pipeline (research → synthesis → refinement → presentation) was executed in approximately 45 minutes of active time. The combination of free and freemium tools made the process accessible to students without budget constraints. Total cost: \$0.

3.5.3 Observed Bottlenecks

- QuillBot’s 125-word limit required splitting longer paragraphs into multiple paraphrase operations.
 - SlidesAI.io’s template library limited design flexibility for specialized content (e.g., mathematical formulas, detailed diagrams).
 - Perplexity’s free tier returned 3–5 sources per query; premium versions would provide more comprehensive coverage.
-

3.6 Pipeline Effectiveness

Running multiple tools in sequence worked. Perplexity gathered credible research. Claude synthesized it into a coherent narrative. QuillBot showed how changing writing style affected clarity. SlidesAI.io generated slides quickly. No single tool did everything. The value came from combining them. Prasad made this point: future AI practitioners need to be problem solvers who can combine multiple tools and disciplines.

4 Conclusions

The pipeline worked. We ran four free tools in sequence, 45 minutes of work, and produced a complete knowledge artifact on Privacy and Data Protection in AI Systems. This is a topic Prasad said matters for responsible AI development.

4.1 Key Accomplishments

The pipeline delivered a complete, multi-format output in 45 minutes using only free and freemium tools. Claude was the strongest link, producing clear, well-structured

writing that engineering students could read. Each tool brought something different to the work: Perplexity’s citations, Claude’s analysis, QuillBot’s style options, SlidesAI.io’s design. That variety mattered.

4.2 Limitations and Trade-Offs

The free tier of each tool imposed practical constraints. QuillBot’s 125-word paraphrase limit required multiple passes for longer text. SlidesAI.io’s template library, while professional, offered limited flexibility for specialized content. Perplexity’s search depth, without premium subscription, returned a smaller citation pool than would be ideal for a comprehensive literature review. These constraints are not failings of the tools themselves but rather reflect the reality that production-quality outputs often require either paid subscriptions or manual refinement.

4.3 Broader Lessons on Human-AI Collaboration

This project illustrates three lessons that directly connect to Prasad’s seminar themes:

1. **Interdisciplinary Orchestration:** Just as Prasad called for engineers, lawyers, ethicists, and social scientists at the “round table” during AI governance, the tool pipeline showed that different AI tools serve different roles. No single tool was sufficient; value emerged from composition.
2. **Transparency and Prompt Engineering:** By documenting every prompt, configuration, and output, this project followed Prasad’s insistence that students “disclose your prompt engineering.” Documentation enables reproducibility. It also trains practitioners to think critically about the inputs and parameters that shape AI outputs. Show the work, not just the result.
3. **Human Judgment and Oversight:** While the tools accelerated the creation process, human decisions were essential at every stage: evaluating research quality, assessing synthesis accuracy, choosing which style variant was most appropriate, and determining when an output was ready for dissemination. The tools amplified human capability; they did not replace it.

4.4 Implications for Privacy in AI

Privacy and data protection in AI are urgent. The EU AI Act is now in effect. Students and engineers entering the field in 2026 will work in environments where privacy is not optional. This project practiced skills that matter: pulling together

regulatory requirements and technical safeguards, then explaining them to different audiences. Modern AI tools let practitioners move beyond purely technical work into broader roles that integrate law, governance, and social impact.

4.5 Future Work

Possible extensions of this project include:

- Integration of additional tools (e.g., Gemini for longer synthesis, Canva for custom presentation design) to compare quality across different providers.
- A longitudinal comparison: rerunning this pipeline in 12 months to assess how evolving AI capabilities and regulatory changes affect the outputs.
- Peer review: having the synthesized content reviewed by privacy experts to assess not just stylistic quality but factual accuracy and completeness.
- Application to other seminar topics: demonstrating that this methodology is generalizable across different subject domains.

4.6 Final Reflection

Prasad closed with "lead with integrity, not just innovation." We tried to live that in three concrete ways. We were transparent about using AI tools, showing the prompts and outputs. We grounded the work in real privacy failures—not hypothetical harms. And we made human judgment central at every step, not as an afterthought.

The tools were powerful and fast. But a tool is just a tool. The hard part is knowing what to ask for and when to say no to what it offers. That judgment is where humans still matter.

5 References

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A Attachments

A.1 Claude AI Output

Privacy and Data Protection in AI Systems

Foundations, Risks, Regulation, and Engineering Safeguards

A. Introduction: Why Privacy Is Critical Now

Artificial intelligence systems deployed between 2025 and 2026 operate at a scale of data collection and inference that earlier software architectures never approached. Where a traditional database stores what a user explicitly typed, a modern AI system can infer what a user implicitly meant, felt, or belongs to — often from data the user never intended to disclose. This shift from explicit data processing to predictive inference is the central reason privacy engineering has moved from a compliance afterthought to a core design constraint.

For engineering students, the implication is direct: privacy is no longer something legal teams bolt on after a model ships. It is an architectural property, like fault tolerance or security, designed into data pipelines and deployment infrastructure from the first commit. A system that performs well on accuracy benchmarks but leaks training data through a poorly secured API has failed at engineering, not merely at ethics. This report synthesizes the major categories of AI privacy risk, the regulatory frameworks governing AI deployment, the technical safeguards available to engineers, and documented incidents that illustrate the cost of getting privacy wrong.

B. Privacy Risks: Real Cases and Technical Vulnerabilities

B.1 Mass Collection and Re-identification

Modern AI pipelines are trained on heterogeneous datasets drawn from logs, images, and device telemetry. The aggregation of these sources across time and platform increases the likelihood that records considered “anonymized” in isolation become re-identifiable once combined. Children and other vulnerable populations face disproportionate exposure, since many consumer platforms collect granular behavioral data without mechanisms for meaningful parental or individual consent.

B.2 Pervasive Surveillance and Function Creep

AI substantially lowers the cost of face and voice recognition, behavioral profiling, and continuous monitoring. A recurring pattern — sometimes called function creep — is that systems built for benign purposes such as analytics or personalization are later repurposed for surveillance or enforcement without new safeguards being added. When governmental or corporate actors adopt algorithmic decision tools at scale, the risk to civil liberties, due process, and democratic oversight increases unless governance keeps pace with deployment.

B.3 Sensitive Inference, Profiling, and Discrimination

Machine learning models can infer sensitive attributes — health status, political affiliation, sexual orientation, or risk scores — from seemingly innocuous inputs. These inferences are frequently inaccurate or stigmatizing, and can exclude individuals from housing, credit, employment, or judicial outcomes. Because biased training data amplifies existing social

inequalities, a model that appears statistically accurate in aggregate can still produce systematically unfair outcomes for specific subpopulations.

B.4 Model-Targeted Attacks and Data Extraction

From an engineering standpoint, this is the risk category closest to traditional security work. Techniques such as model inversion, membership inference, and adversarial prompting can extract training data directly from a deployed model or reveal whether a specific individual's data was used during training. Poorly protected model endpoints and loosely governed third-party integrations expand the attack surface considerably, turning an inference API into an unintended data exfiltration channel.

B.5 Insecure Data Sharing and Supply-Chain Exposure

AI development is rarely a single-organization activity. Data brokers, cloud vendors, and third-party annotators all touch personal data at some stage of the pipeline. Weak contractual and technical controls across this supply chain allow personal data to leak well outside the boundaries an end user would assume, and cross-border data flows complicate accountability further when legal regimes differ across jurisdictions.

B.6 Documented Incidents

Several public incidents illustrate how these abstract risk categories materialize in practice:

- Clearview AI compiled billions of facial images scraped from the public web without consent, building a commercial face-recognition product that triggered regulatory enforcement and litigation across multiple jurisdictions — a clear case of unlawful mass collection feeding a biometric surveillance tool.
- A 2023 software bug in a widely used generative AI chat product briefly exposed some users' chat titles and partial billing information, demonstrating that AI products inherit conventional data-breach risks in addition to model-specific ones.
- Generative models have repeatedly shown the capacity to retain and, under certain prompting conditions, reveal fragments of their training data — a recurring concern that connects directly to the model-inversion risks described above.
- Facial recognition deployments in public and commercial settings have raised recurring privacy concerns when used for identification or surveillance without adequate legal basis, transparency, or oversight mechanisms.

Across these cases, three lessons recur: harm rarely originates from the model alone but from weak data governance and vendor oversight surrounding it; biometric, health, and children's data warrant categorically stricter controls; and the absence of testing for memorization, leakage, and re-identification before deployment is itself a design failure, not bad luck.

C. Regulatory Landscape: GDPR and the EU AI Act

C.1 GDPR as the Baseline Framework

The General Data Protection Regulation remains the primary legal framework governing lawful processing of personal data in the EU and, by extension, much global AI practice that touches EU residents' data. Its core principles — lawful basis for processing, purpose limitation, data

minimization, and individual rights such as access, correction, and erasure — apply directly to AI training pipelines, which frequently ingest personal data at a scale that makes individual consent difficult to operationalize.

C.2 The EU AI Act: AI-Specific Obligations

The EU AI Act complements GDPR by adding AI-specific duties rather than replacing existing data-protection law. Its key mechanisms include:

- Risk-based classification — systems are categorized as unacceptable, high, limited, or minimal risk, with high-privacy-impact uses such as indiscriminate biometric surveillance banned or tightly restricted.
- Data governance for high-risk systems — providers must maintain risk-management systems, technical documentation, and operational logs, and demonstrate dataset quality, directly limiting harms from biased training data.
- Transparency requirements — providers must disclose capabilities, intended use, and limitations, supporting GDPR's fair-processing and informed-consent principles.
- Human oversight mandates — high-risk systems require oversight proportionate to risk, reducing unreviewable automated decisions.
- Robustness and cybersecurity requirements — providers must show resilience against adversarial manipulation, addressing the extraction risks in Section B.4.
- Logging and post-market monitoring — obligations that intersect with GDPR breach-notification regimes.
- Conformity assessment and independent audit — high-risk systems must be verified before market placement.

For engineering teams, the practical consequence is that compliance now requires coordinated effort across legal and technical functions: GDPR governs the lawful basis and individual rights tied to personal data, while the AI Act governs the documentation, risk management, and technical robustness of the system processing that data. Detailed logging and model cards, increasingly required under the Act, also directly support the kind of Data Privacy Impact Assessments that good engineering practice already recommends.

D. Technical Safeguards: Privacy by Design, Differential Privacy, and Federated Learning

D.1 Privacy by Design

Privacy by Design (PbD) is the principle of embedding privacy protections into system architecture from the outset, rather than retrofitting them after deployment. Applied to AI, this means data minimization and purpose limitation at the collection stage, privacy-preserving training techniques such as on-device or federated learning, cautious and tested de-identification, hardened model endpoints with authentication and anomaly monitoring, restricted inference scope that avoids predicting sensitive attributes unless strictly justified, and default privacy settings that protect users without requiring active configuration. Crucially, PbD also requires full-lifecycle protection — privacy controls must follow data from collection through training, deployment, maintenance, and eventual deletion, not just at the point of intake.

D.2 Differential Privacy

Differential privacy is a mathematical guarantee that bounds how much any single output can reveal about any one individual's contribution to a dataset. In practice, engineers implement it by injecting calibrated noise into gradients, counts, or aggregated statistics during training or analytics. This materially reduces the risk that a model memorizes specific records and makes membership inference — determining whether a particular person's data was used in training — substantially harder. It also gives engineering teams a quantifiable privacy budget, replacing vague policy assurances with a tunable parameter. The tradeoff is real: stronger privacy guarantees typically reduce model accuracy or utility, so teams must measure that tradeoff explicitly rather than treating it as a fixed cost.

D.3 Federated Learning

Federated learning trains a shared model across many devices or institutions without centralizing raw data. Each client computes a local update and transmits only that update — not the underlying personal records — to a central server for aggregation. This is particularly valuable in sensitive domains such as healthcare, mobile personal assistants, and any setting where institutional or regulatory constraints make centralizing raw data impractical. It is important to note, however, that federated learning alone does not guarantee privacy: model updates themselves can still leak information about the underlying data through gradient inversion or related techniques.

D.4 Layered Protection: Combining Techniques

The strongest engineering pattern combines these safeguards rather than relying on any single one. Federated learning reduces the movement of raw data; differential privacy reduces what can be inferred from the updates or the resulting model; and both are typically paired with encryption, secure aggregation, strict access controls, and continuous monitoring. A useful mental model for students: a health application might train locally on each patient's device using federated learning, then apply differential privacy to the shared updates so that even the central aggregating server cannot determine which patient contributed which signal.

These techniques should be selected based on where the dominant risk sits. Differential privacy is most relevant when the primary concern is leakage from training data or model outputs; federated learning is most relevant when raw data must remain local for regulatory, institutional, or trust reasons; and both together are warranted when data is highly sensitive and the model spans many users or sites.

E. Governance, Human Oversight, and Interdisciplinary Collaboration

E.1 Transparency as an Engineering Deliverable

Transparency is frequently framed as a legal or ethical obligation, but it is equally an engineering deliverable. Model cards, data provenance records, and clear user-facing notices are artifacts that engineering teams produce alongside the model itself. They make Data Privacy Impact Assessments possible, give auditors a basis for verification, and give end users enough information to exercise rights such as correction or deletion. A model without this

documentation is, from a governance standpoint, no more trustworthy than undocumented code is maintainable.

E.2 Human-in-the-Loop as a Privacy Control

Human oversight is not merely a regulatory checkbox required for high-risk systems under the EU AI Act; it functions as a direct privacy control. Meaningful human review of high-stakes automated decisions — with clear escalation and appeal pathways — catches the kind of inaccurate sensitive-attribute inferences described in Section B.3 before they cause irreversible harm. This is a collaborative relationship rather than a constraint imposed on AI capability: well-designed human-AI collaboration treats the human reviewer as a check on inference quality, not a rubber stamp on automated output.

E.3 Why This Requires Interdisciplinary Teams

No single discipline can deliver privacy-respecting AI alone. ML engineers tune the technical safeguards in Section D; legal specialists translate GDPR and AI Act obligations into requirements; security teams harden endpoints against extraction attacks; and product teams design defaults and consent flows users can actually understand. A gap in any one area reintroduces risk even when the others function well — sound differential privacy does not compensate for a vendor contract with no privacy clauses, and a thorough privacy policy does not compensate for an unauthenticated model API.

E.4 Practical Implications for AI Engineers

For students entering AI engineering roles, several practices follow directly from the risks and safeguards discussed above:

- Treat data minimization as a default design constraint — collect only what the task requires.
- Run a Data Privacy Impact Assessment at project initiation, before architecture decisions are finalized.
- Apply differential privacy or federated learning wherever training data is personal or sensitive, and document the accuracy tradeoff.
- Harden every model-serving endpoint against extraction attacks with authentication, rate limiting, and anomaly monitoring.
- Produce model cards and provenance documentation as a deliverable alongside the trained model, not separate paperwork.
- Build human review and appeal mechanisms into pipelines that produce consequential decisions about individuals.
- Audit vendors and cloud dependencies for privacy practices with the same rigor applied to in-house systems.

Conclusion

Privacy in AI systems is not a constraint layered on top of capability; it is part of what makes a system fit for deployment at all. The risks reviewed here — mass collection, surveillance creep, biased inference, model extraction, and supply-chain leakage — are well documented and recurring, not speculative. The regulatory response, through GDPR and the EU AI Act, has converged on a consistent set of expectations: minimize data, document everything, secure the

model, and keep a human in the loop for consequential decisions. The technical safeguards — privacy by design, differential privacy, and federated learning — give engineers concrete tools to meet those expectations rather than treating them as externally imposed burdens. The students who will build the next generation of AI systems are best served by internalizing this early: a model that cannot demonstrate how it protects the people whose data it learned from has not finished being engineered.

Sources synthesized from peer-reviewed and institutional research on AI ethics, data protection law, and privacy-preserving machine learning (2023–2026).

A.2 SlidesAI.io Output

The Main Problem

The Core Objective
The fundamental challenge for AI engineers is to prevent the collection, exposure, or misuse of personal data throughout the pipeline.

Risk Phase	Primary Concern
Collection	Gathering data without consent or beyond the necessary scope.
Processing	Models learning and replicating sensitive personal patterns.
Deployment	Unauthorized access or "membership inference" attacks on live models.

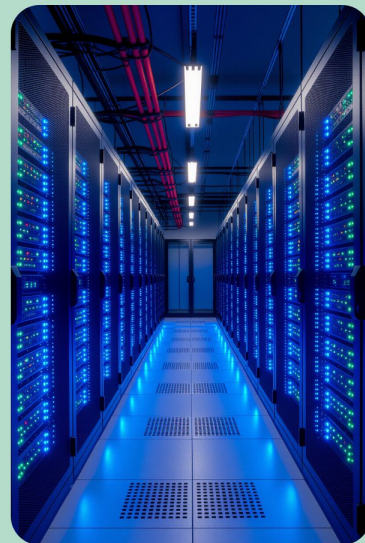
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Why Privacy Matters Now

The Scale of AI Inference

Modern AI systems differ from traditional software in how they handle information:

- 1 **Massive Data Aggregation:** AI requires vast datasets, increasing the "blast radius" of any potential leak.
- 2 **Sensitive Inferences:** Models can predict private traits (health, orientation, politics) from seemingly non-sensitive data.
- 3 **Persistent Memory:** Generative models may "memorize" training data, leading to accidental exposure of personal details long after training ends.



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Key Privacy Risks

Critical Vulnerabilities in AI



Re-identification

Anonymized datasets can often be "de-anonymized" by cross-referencing with other AI-driven sources.



Surveillance Creep

Tools built for one purpose (e.g., security) being repurposed for invasive monitoring.



Supply-Chain Exposure

Using third-party pre-trained models or datasets introduces "hidden" privacy risks.



Algorithmic Discrimination

Biased data leads to discriminatory outcomes that violate individual rights and privacy.

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Real Incidents and Evidence

Documented Privacy Failures



Scraping Practices: Clearview AI's use of billions of social media photos without user knowledge or consent.



Data Memorization: Research showing LLMs can be prompted to reveal credit card numbers or addresses found in training sets.



Facial Recognition: High error rates and invasive tracking in public spaces leading to city-wide bans.



Generative Leakage: Image generators producing recognizable faces of private individuals from the training corpus.

Source: Documented Industry Case Studies

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Regulatory Landscape

Global Standards for AI

GDPR

Focuses on "**Lawful Processing**" and the "**Right to Explanation**" for automated decisions.

EU AI Act

Classifies AI systems by risk level (Unacceptable, High, Limited, Minimal).



Key Requirements

- **Transparency:** Users must know they are interacting with an AI.
- **Human Oversight:** Critical decisions must have a "**human in the loop**."
- **Audits:** High-risk systems require rigorous third-party testing.

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Engineering Safeguards

Privacy by Design



Data Minimization

Only collect the specific data points required for the model's task.



Secure Endpoints

Protecting APIs to prevent "model inversion" attacks where attackers reconstruct training data.



Lifecycle Controls

Implementing strict data deletion policies and "unlearning" capabilities for models.



Default Protections

Systems should be private by default, requiring no user action to secure their data.

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Differential Privacy and Federated Learning

Differential Privacy

Adds mathematical noise to data to mask individual identities.

Federated Learning

Trains models locally on devices; only weights are shared.



The Balance

Accuracy vs. Privacy vs. Compute.

DP Tradeoffs

Provides strong guarantees but can reduce model accuracy if the **"noise"** is too high.

FL Tradeoffs

Keeps data on-device, but requires complex synchronization and remains vulnerable to certain gradient attacks.

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Governance and Human Oversight

The Human Element



Model Cards

Standardized documents explaining a model's training, risks, and intended use.



Data Provenance

Maintaining a clear "paper trail" of where data came from and how it was cleaned.



DPIAs

Formal reviews conducted before a system is deployed to identify potential harms.



Interdisciplinary Teams

Collaboration between engineers, ethicists, and legal experts.

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Practical Takeaways

The Engineer's Checklist

- ✓ 1. **Audit your data:** Know exactly what personal information is in your training set.
- ✓ 2. **Minimize at the source:** If you don't need the data, don't collect it.
- ✓ 3. **Test for leakage:** Use adversarial attacks to see if your model "leaks" training data.
- ✓ 4. **Prioritize Transparency:** Be clear with users about how their data influences the AI.

Conclusion

Privacy is not a barrier to innovation—it is a prerequisite for sustainable, trustworthy AI deployment.



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